Example 15.5

The transfer function is

$$h(w) = \sum_{j=-\infty}^{\infty} b_j e^{-iwj} \qquad b_0 = \tfrac{1}{2};\ b_1 = \tfrac{1}{2};$$

$$b_j = 0 \text{ otherwise.}$$

$$= \frac{1}{2} + \frac{1}{2} e^{-iw}$$

The squared gain is:

$$|h(w)|^2 = h(w)\,\overline{h(w)}$$

$$= \frac{1}{2}(1+e^{-iw}) \times \frac{1}{2}(1+e^{iw})$$

$$= \frac{1}{4}(2+e^{iw}+e^{-iw})$$

$$= \frac{1}{4}(2+2\cos w)$$

$$= \frac{1}{2}(1+\cos w)$$

$$g_1(w) = |h(w)|^2 g_0(w)$$

$$= |h(w)|^2 \frac{\sigma_z^2}{2\pi}$$

$$= \frac{\sigma_z^2}{4\pi}(1+\cos w)$$

$$g_2(\omega) = |h(\omega)|^2 g_1(\omega)$$
$$= |h(\omega)|^2 \left[|h(\omega)|^2 g_0(\omega) \right]$$
$$= |h(\omega)|^4 g_0(\omega)$$
$$= \frac{\sigma_z^2}{2\pi} |h(\omega)|^4$$

Generally $g_k(\omega) = |h(\omega)|^2 g_{k-1}(\omega)$

$$= |h(\omega)|^{2k} g_0(\omega)$$
$$= \frac{\sigma_z^2}{2\pi} |h(\omega)|^{2k}$$
$$= \frac{\sigma_z^2}{2\pi} \left[|h(\omega)|^2 \right]^k$$
$$= \underbrace{\frac{\sigma_z^2}{2\pi}}_{g_0(\omega)} \underbrace{\left[\frac{1}{2}(1+\cos\omega) \right]^k}_{\text{Squared gain wrt a transformation from } y_t^{(0)} \to y_t^{(k)}}$$
$$= \frac{\sigma_z^2}{2\pi} \left[\underbrace{\frac{1}{2^{k/2}}(1+\cos\omega)^{k/2}}_{\tilde{h}(\omega)} \right]^2$$

$\{ y_t^{(k)} \}$ is a transformation of $\{ y_t^{(0)} \}$

with transfer function

$$\tilde{h}(\omega) = \frac{1}{2^{k/2}}(1+\cos\omega)^{k/2}$$

$$g_k(\omega) = |\tilde{h}(\omega)|^2 g_0(\omega)$$